Innovative Fluorinated Boron-Based Electrolyte Harmonizes Interface Chemistry-Electrochemistry of Mg Anode for Accessible Rechargeable Mg Battery

Xiaochen Liu[a], Jiaxin Wen[b], Xingwang Zhao[a], Bo Shang[a], Jinglei Lei[a,*], Lingjie Li[a,c,*], and Fusheng Pan[c]

[a] *School of Chemistry and Chemical Engineering, Chongqing University, Chongqing, 400044, P.R. China*

[b] *Chongqing Industry Polytechnic College, Chongqing, 401120, P.R. China*

[c] *National Engineering Research Center for Magnesium Alloys, Chongqing University, Chongqing, 400044, P.R. China*

*Corresponding authors. E-mail address: jllei@cqu.edu.cn (J. Lei), ljli@cqu.edu.cn (L. Li)

ABSTRACT

The development of promising boron-based electrolyte for rechargeable magnesium batteries (RMBs) is constrained by high costs and cumbersome synthesis. Herein, an accessible and innovative magnesium phenyl fluoroborate complex electrolyte (MPFBC) has been designed, which employs cheap boron trifluoride as both the boron and fluorine source, utilizing a facile Lewis acid-base reaction for synthesis. The MPFBC electrolyte is capable of stable cycling of Mg||Mg cells at 0.5 mA cm^{-2} with low overpotential (<100 mV) and maintains a decent discharge specific capacity of 62.6 mAh g^{-1} after 1000 cycles of the Mg||Mo_6S_8 battery at 0.5 C. Further, the complementary advantages of chemistry and electrochemistry have been highlighted for the first time to harmonizes interface chemistry of Mg anode. A complementation combining the in-situ pre-corrosion driven by chemical process and the Mg^{2+}-conducting layer formed through electrochemical process enables controllable interface chemistry and interface evolution. The design and synthesis of MPFBC pave a feasible way for commercialization of boron-based electrolytes, and simultaneously the interface regulation strategy of chemistry-electrochemistry complementary advantages enriches the interface theories of RMBs.

Keywords: Rechargeable magnesium batteries, Boron-based electrolyte, Interface chemistry, B & F-containing interface, Complementary advantages

1. Introduction

The application of electrochemical energy storage technologies can effectively mitigate the intermittency issues of renewable energy, enhance the efficiency of energy utilization, and facilitate the transition towards a low-carbon energy structure[1,2]. Rechargeable magnesium batteries (RMBs) have emerged as one of the premier choices for large-scale electrochemical energy storage, owing to the utilization of magnesium metal anode which boasts abundant reserves, high theoretical energy density (3833 mAh cm^{-3}; 2205 mAh g^{-1}), and a low reduction potential (−2.37 V *vs.* SHE) [3,4]. However, the advancement of RMBs is hindered by the limitations in electrolyte progress, primarily due to two key factors: the lack of electrolytes that can balance performance and cost, and the insufficiency of guidelines on tuning the electrolyte/electrode interface [5]. Therefore, the design of innovative and economical electrolytes in conjunction with interface regulation is imperative and challenging [6-8].

Subsequent to the pioneering showcase of magnesium plating/stripping facilitated by Grignard reagents, considerable endeavor has been undertaken and promising progress has been achieved in aprotic electrolytes for RMBs over the past two decades [9,10]. Among these, boron-based electrolytes featuring boron-center anions have progressively emerged as one of the most viable candidates [11-13]. This is attributable to the rich diversity of boron compounds and their unique chemical properties, which provide broad prospects for the development of functionalized boron-based electrolytes [14,15]. The most representative example is magnesium tetrakis(hexafluoroisopropyloxy)borate electrolyte ($Mg[B(hfip)_4]_2$), which integrates the merits of high anode stability, decent conductivity, excellent compatibility and reversible Mg^{2+} deposition/dissolution [16]. Similar boron-based electrolytes also include $Mg[B(Otfe)_4]_2$ [17,18] and Mg-FP [19]. The anions in these electrolytes, which commonly feature boron centers bonded to bulky and electron-withdrawing fluoroalkoxy groups, can react with the Mg anode to form a stable solid electrolyte interphase (SEI) [20]. Other studies have revealed that asymmetric boron-center anions with lower LUMO energy levels can induce the formation of robust boron-containing interfaces, leading to enhanced Mg^{2+} diffusion and improved cycling stability [21,22]. Although boron-center anions are adept at constructing designable interfaces to promote battery performance, their substantial cost and complex synthesis procedure pose a limitation to widespread application in RMBs [18,23]. To address these challenges, the resolution entails employing low-cost raw materials and streamlining the synthetic routes to develop novel boron-based electrolytes.

Furthermore, the impact of the interface reactions between the electrolyte and electrode on the interface chemistry and electrochemical performance demands much concern. The interface reactions primarily consist of chemical and electrochemical reactions, which occur during the resting process and the charge-discharge process of battery operation, respectively [24]. Continuous interface interactions can engender dynamic alterations in the interface chemistry, ultimately determining key performance metrics such as cycle life, safety, and coulombic efficiency [25,26]. Rational utilization of interface reactions can adjust the interface chemistry to enhance the overall electrochemical performance of the batteries [27,28]. However, the role of chemical reactions tends to be overshadowed by a predominant focus on electrochemical reactions, leading to fundamental gaps in understanding how chemical reactions of electrolyte components affect interface chemistry. The reason may be that the resting process precedes the charging-discharging process and has a relatively short duration. This imbalance is prejudicial to the development of interfacial theories. Therefore, it is of practical significance to ascertain the influence of electrolyte components on chemical reactions and even to empower interface chemistry by harmonizing chemical and electrochemical reactions.

Herein, we have successfully developed an innovative and accessible magnesium phenyl fluoroborate complex electrolyte (MPFBC) through a facile Lewis acid-base reaction between cheap reactants: BF_3 and PhMgCl. This innovation substantially reduces the cost and simplifies the preparation of boron-based electrolytes. The MPFBC can ensure stable cycling of Mg||Mg cells at 0.5 mA cm^{-2} with low overpotential (<100 mV) and retains a satisfactory discharge specific capacity (62.6 mAh g^{-1}) after 1000 cycles of the Mg||Mo_6S_8 battery at 0.5 C. Furthermore, the interface chemistry and electrode process of Mg anode in MPFBC have been thoroughly investigated, with a special focus on the cells in open circuit (experienced chemical process) and after cycling (experienced electrochemical process). Mg is passivated by the corrosion reaction during the chemical process, while the execution of electrochemical process can activate Mg by SEI construction. In turn, the corrosion morphologies initiated by chemical reactions benefit the relatively uniform Mg stripping/plating in the subsequent cycles — an absent capability in purely electrochemical reactions. Utilizing the complementary advantages of the in-situ pre-corrosion driven by chemical reactions and the Mg^{2+}-conducting SEI formed through electrochemical reactions enables controllable interface chemistry and interface evolution. This work provides an accessible boron-based electrolyte with a promising outlook for large-scale commercialization, and contributes a practical and universal interface regulation strategy for RMBs.

2. Results and discussion

Boron trifluoride (BF_3) as the simplest and cheapest boron compound was firstly applied to electrolytes decades ago, but reversible magnesium deposition/dissolution failed to be achieved due to the lack of effective electroactive species [29]. Then it has remained out of the spotlight for an extended period. Interestingly, BF_3 has a similar molecular structure to aluminum chloride ($AlCl_3$), a Lewis acid used for preparing all-phenyl complex (APC) electrolyte [30]. Inspired by this, substituting $AlCl_3$ with BF_3 to react with phenylmagnesium chloride (PhMgCl) may be a facile routine to obtain boron-based electrolytes (Figure 1a). According to the hybrid orbital theory, the central B and Al atom adopt sp^2 hybridization and the empty p-orbital is perpendicular to the hybrid orbitals, which endow BF_3 and $AlCl_3$ with a planar triangular geometry and property of electron acceptor [31]. The electrostatic potential (Figure 1b) shows that the positive charges are concentrated around the central Al and B atom, indicating local electrophilicity. The dual descriptor potential (Figure 1c&d) and other local reactivity descriptors (Figure 1e) further demonstrate that the central atoms are more susceptible to nucleophilic attacks [32,33]. From the view of charge distribution and reactivity prediction, BF_3 is indeed an electrophilic Lewis acid and has potential to replace $AlCl_3$ in reactions with nucleophilic Lewis bases to synthesize novel boron-based electrolytes. Moreover, BF_3 is a high-quality boron source and fluorine source which provides material basis for the construction of a boron & fluorine-containing interface by chemical and electrochemical reaction.

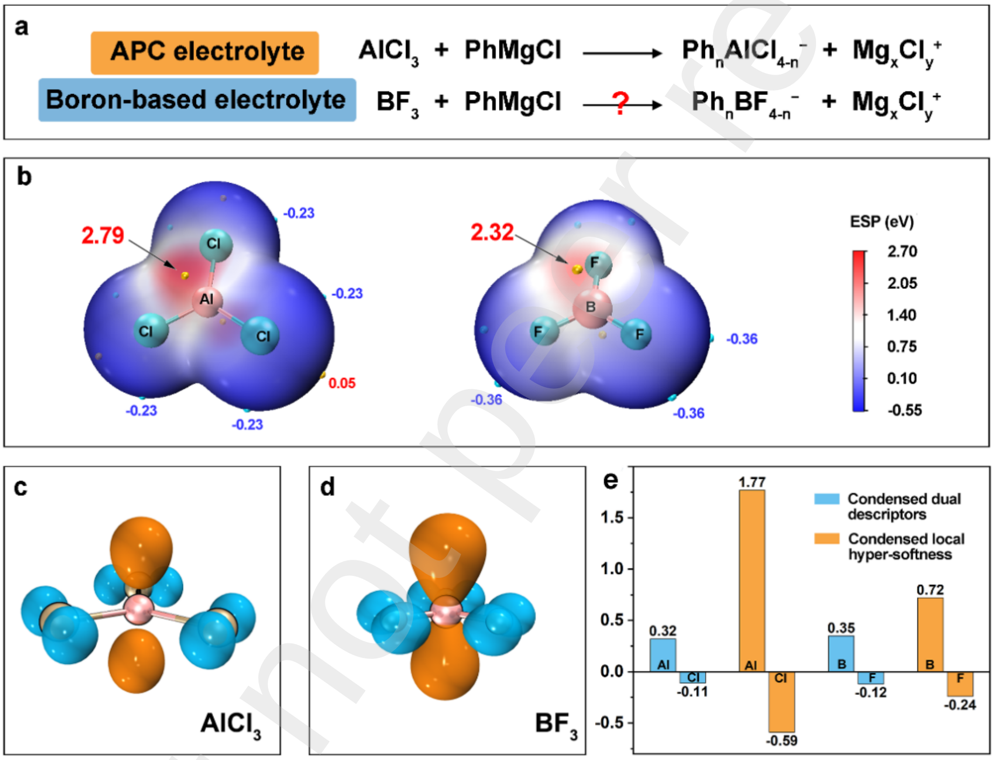

Figure 1. The basis for electrolyte design. (a) Synthesis scheme for APC and boron-based electrolytes. (b) Electrostatic potential of $AlCl_3$ and BF_3. Significant surface local maxima and minima are represented as orange and cyan spheres with labels. Dual descriptor potential of (c) $AlCl_3$ and (d) BF_3. The orange lobes represent positive phase suggesting electrophilic proclivity, while the cyan lobes represent negative phase suggesting nucleophilic proclivity. (e) Local reactivity descriptors calculated based on conceptual density functional theory. The atoms with positive values

tend to accept electrons, while the atoms with negative values tend to donate electrons. The unit for condensed dual descriptors is "e" and the unit for condensed local hyper-softness is "e hartree^{-2}".

A convenient synthesis scheme by reaction between PhMgCl and BF_3 can yield magnesium phenyl fluoroborate complex electrolyte (MPFBC) without harsh reaction conditions and lengthy purification. Theoretically, the phenyl group from PhMgCl launches a nucleophilic attack onto the central atom of BF_3, thus forming the boron-center anions. In practice, the complex reaction mechanism and product identification are further characterized. A basic group of MPFBC is named B1P1M0, which is composed of 0.25 M BF_3-PhMgCl in THF. The negative mode of electrospray ionization mass spectrometry (ESI-MS) as shown in Figure 2a, the peaks with m/z ratio of 87.00 and 319.16 are attributed to BF_4^- and $[(C_6H_5)_4B]^-$, respectively. The formation of these two boron-center anions indicates that F^- and Ph^- have undergone the ligand exchange reactions during the Lewis acid-base reactions. A similar phenomenon can also be observed in the APC electrolyte [34]. The remaining mass spectrometry signals are related to the reaction products between boron-center anions and methanol (the mobile phase in the test). The peak located at 151.05 m/z ratio corresponds to $[(CH_3OH)_2BF_4]^-$, a complex formed by the interaction of BF_4^- with CH_3OH. Similarly, the peak at 273.14 m/z ratio is assigned to $[(C_6H_5)BF_3(CH_3OH)_4]^-$, indicating the presence of the third boron-center anion, $[(C_6H_5)BF_3]^-$. Specially, the species with m/z ratio of 259.13 represents $[(C_6H_5)_3BOH]^-$, which is the product of hydrolysis reaction between the fourth boron-center anion $[(C_6H_5)_3BF]^-$ and trace of H_2O in methanol. It should be pointed out that due to different solubility of four anions in methanol, the intensity of mass spectrum signals cannot be employed to compare the relative concentration of anions in electrolyte.

Nuclear magnetic resonance (NMR) further confirms the existence of four boron-center anions. As depicted in Figure 2c, the reactant BF_3 exhibits two signals at −0.58 and −1.46 ppm in ^{11}B NMR spectrum, which correspond to free BF_3 and the complex $BF_3 \cdot THF$. In B1P1M0 electrolyte, most of BF_3 is converted into BF_4^- (−1.56 ppm), with only a minor fraction of free BF_3 remaining through equilibrium reactions. The other three weak peaks located at the low field region are expected to be the anions containing phenyl groups. The boron atom in anion is influenced by the delocalization effect of phenyl group, leading to a decrease in electron density and an increase in chemical shift. The higher degree of phenyl substitution on boron is, the more positive chemical shift becomes. Therefore, the peaks at 18.71, 28.01, and 44.16 ppm are identified as $[(C_6H_5)BF_3]^-$, $[(C_6H_5)_3BF]^-$, and $[(C_6H_5)_4B]^-$, respectively. In ^{19}F NMR, the reactant BF_3 exhibits two signals at −153.0 and −157.2 ppm corresponding to free BF_3 and the complex $BF_3 \cdot THF$. In B1P1M0, three peaks around −153.4, −147.5 and −141.1 are related to BF_4^-, $[(C_6H_5)BF_3]^-$ and $[(C_6H_5)_3BF]^-$. Consequently, the following reaction formula is used to describe the formation of MPFBC:

$$BF_3 + PhMgCl \longrightarrow BF_4^- + PhBF_3^- + Ph_3BF^- + Ph_4B^- + Mg_xCl_y^+$$

The structure and electrostatic potential of four boron-center anions are intuitively illustrated in Figure 2e. The dispersibility of the negative charges enhances as the number of phenyl groups on the boron atom grows. In BF_4^-, lacking a phenyl group, the negative charges are concentrated on the F atom. Whereas in $[(C_6H_5)_4B]^-$, which contains four phenyl groups, the negative charges are extensively dispersed through delocalization effects. The high electron delocalization characteristic can weaken the electrostatic interaction between the anion and the cation [35]. In other words, phenyl-substituted boron-center anions are less prone to forming ion pairs with Mg^{2+} compared to BF_4^-.

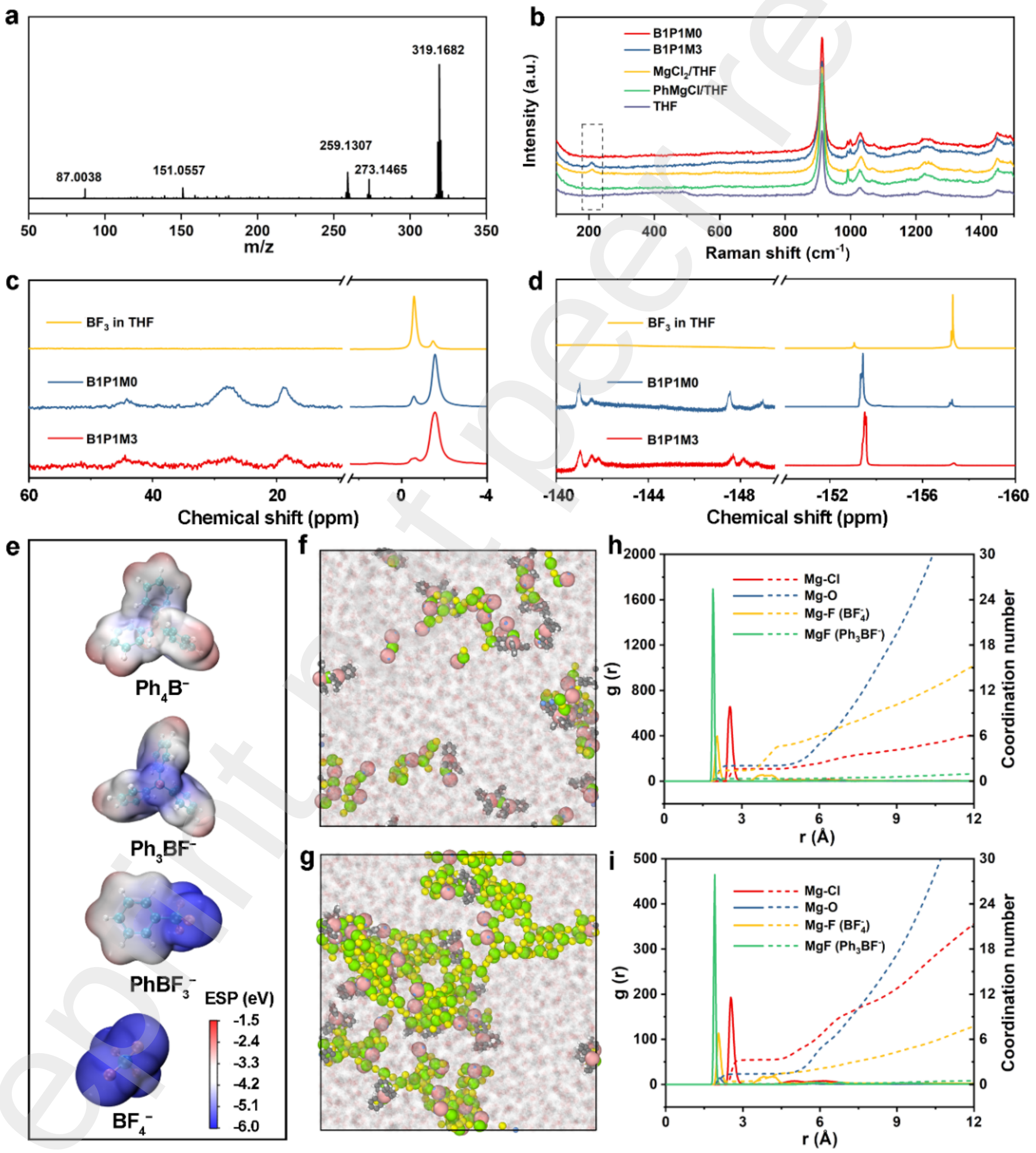

Figure 2. Composition and solvation structures of MPFBC. (a) ESI-MS of the B1P1M0 in negative mode. (b) Raman spectra of electrolytes and reactants. (c) ^{11}B NMR (d) ^{19}F NMR spectrum of electrolytes and BF_3. (e) Electrostatic potential of Ph_4B^-, Ph_3BF^-, $PhBF_3^-$ and BF_4^- anions. The snapshots and RDFs from MD simulation trajectories in (f & h) B1P1M0 and (g & i) B1P1M3.

However, B1P1M0 has difficulty in supporting the reversible Mg^{2+} deposition/dissolution, as shown in Figure 3a. To rescue the electrochemical performance, the simplest method is to introduce $MgCl_2$ to B1P1M0, because the additional chloride ions can combine with Mg^{2+} to form electrochemically active species [36]. Obviously, the addition of $MgCl_2$ significantly improves electrochemical behavior of Mg^{2+}, as shown by CV curves of B1P1M3 (0.25 M BF_3-PhMgCl-3$MgCl_2$ in THF) in Figure 3b. Meanwhile, the $MgCl_2$ in B1P1M3 has no influence on the Lewis acid-base reaction, and the composition of boron-center anions is essentially consistent with that of B1P1M0 (Figure S1 and Table S1).

The divergent electrochemical behavior of the aforementioned two electrolytes stems from different solvation structures of Mg^{2+}. The positive mode of ESI-MS provides clues for the existing form of Mg^{2+}, although it cannot fully reflect the actual solvation structures in electrolyte. Mg^{2+} forms ion-pairs with BF_4^- in B1P1M0 (Figure S2 and Table S2). In contrast, Mg^{2+} mainly exists as various Mg^{2+}-Cl^- species rather than $(MgBF_4)^+$ in B1P1M3 (Figure S3 and Table S3). Raman spectroscopy, as illustrated in Figure 2b, offers more reliable evidence through a characteristic peak at 209.1 cm^{-1}, which is attributed to the presence of Mg^{2+}-Cl^- species in B1P1M3, specifically the $[Mg_2Cl_3 \cdot 6THF]^+$ [37]. Furthermore, molecular dynamics (MD) simulation (Figure 2f-i) provides a deep understanding of the solvation structures. Within the first solvation shell of Mg^{2+}, radial distribution functions (RDF) indicate that in B1P1M0, the coordination number of Mg^{2+} with F from BF_4^- is 1.43, whereas with F from Ph_3BF^- is 0.32. The structures of the extracted ion pairs (Figure S4) reveal that Mg^{2+} interacts with two F atoms from BF_4^- and one F atom from Ph_3BF^-. Integrating the above information, it can be deduced that there is one boron-center anion surrounding each Mg^{2+} in the B1P1M0, implying that boron-center anions are fully engaged in the solvation structure of Mg^{2+}. In B1P1M3, a large amount of $MgCl_2$ leads to an increase in the Mg-Cl coordination number to 3.27 (1.62 in B1P1M0), with corresponding decreases in the Mg-F (from BF_4^-) and Mg-F (form Ph_3BF^-) coordination numbers to 0.42 and 0.08, respectively. This implies that Mg^{2+}-Cl^- species become the dominant solvation structures in B1P1M3. MD simulation further corroborates the findings from ESI-MS and Raman spectroscopy. Mg^{2+}-Cl^- species, particularly $[Mg_2Cl_3 \cdot 6THF]^+$, have been extensively studied and confirmed to be an electrochemical active species in chloride-containing electrolytes [29,38], while $(MgBF_4)^+$ species cannot support the electrodeposition of Mg^{2+}. Therefore, $MgCl_2$ plays a key role in altering the solvation structure of MPFBC resulting in the ability of reversible Mg^{2+} deposition/dissolution.

We also prepare B2P1M0 (0.25 M $2BF_3$-PhMgCl in THF) and B1P2M0 (0.25 M BF_3-2PhMgCl in THF) by adjusting the Lewis acid-to-base ratio. As illustrated in Figure S5 and Figure S6, these electrolytes, without the addition of $MgCl_2$, also fail to achieve reversible Mg^{2+} deposition/dissolution. Upon introducing $MgCl_2$ to B2P1M0, the resultant B2P1M3 (0.25 M $2BF_3$-PhMgCl-$3MgCl_2$ in THF) gradually exhibits a mediocre Mg plating and stripping behavior following a conditioning process (Figure 3c). Among all prepared MPFBC samples, B1P1M3 can achieve the maximum anodic current density (10 mA cm^{-2}) and the highest efficiency (99%) of Mg^{2+} dissolution/deposition compared to other samples (Figure S7 and S8). In addition, B1P1M3 keeps decent anodic stability on Mo electrode up to 2.5 V *vs*. Mg/Mg^{2+} (Figure S9), and its ionic conductivity is 49.8 μS cm^{-1} at 30 °C (Figure S10).

Symmetric Mg||Mg cells are employed to evaluate the polarization properties and stability in MPFBC. The best-performing B1P1M3 exhibits a polarization voltage consistently below 100 mV during prolonged cycling under 0.5 mA cm^{-2} and 0.25 mAh cm^{-2} (Figure 3d and Figure S11-S15). When the current density is gradually increased to 2.0 mA cm^{-2}, its polarization voltage remains at merely 284 mV (Figure 3f). Furthermore, asymmetric Mg||SS cells are utilized to assess the coulombic efficiency of Mg^{2+} dissolution/deposition. In Figure 3e, the initial coulombic efficiency of B1P1M3 is 75.66%. After cycling, this value climbs to 99% and maintains this high level consistently in subsequent cycles, demonstrating excellent reversibility. The discharge median voltage reflects an initial rapid decline from 262 mV to 75 mV, followed by a slight increase as cycling proceeded (Figure S16). The coulombic efficiency of B2P1M3 and B1P2M0 is lower and more fluctuating, accompanied by higher discharge median potential (Figure S17 & S18). Regarding rate performance at different current densities (Figure 3g and S19), B1P1M3 still shows reasonable overpotential and coulombic efficiency. These results affirm that B1P1M3 possesses the most favorable electrochemical performance. Hence, we select it as the subject for subsequent research.

The compatibility of B1P1M3 with cathode is the premise for its application in RMBs. The classical intercalation-type Mo_6S_8 cathode is used to assemble RMBs. In long-term cycling tests, the discharge specific capacity of Mg|B1P1M3|Mo_6S_8 reaches 70.6 mAh g^{-1} after an activation process at 0.5 C (Figure S20). In subsequent cycles, the battery shows capacity fluctuation and a slight decrease, while maintaining a high coulombic efficiency close to 100%. After 1000 cycles, the discharge specific capacity still remains at 62.6 mAh g^{-1}. B1P1M3 can support the conversion-type CuS cathode to experience magnesiation and demagnesiation as well (Figure S21). Contrastively, the APC electrolyte is unable to support the normal operation of the CuS||Mg cells (Figure S22). The cost advantage is also an important impetus for the large-scale application of B1P1M3. Figure S23 summarizes the costs of boron-based electrolytes reported in recent years. B1P1M3 is the cheapest one, and its cost is one thirtieth of that of $Mg[B(hfip)_4]_2$ electrolyte.

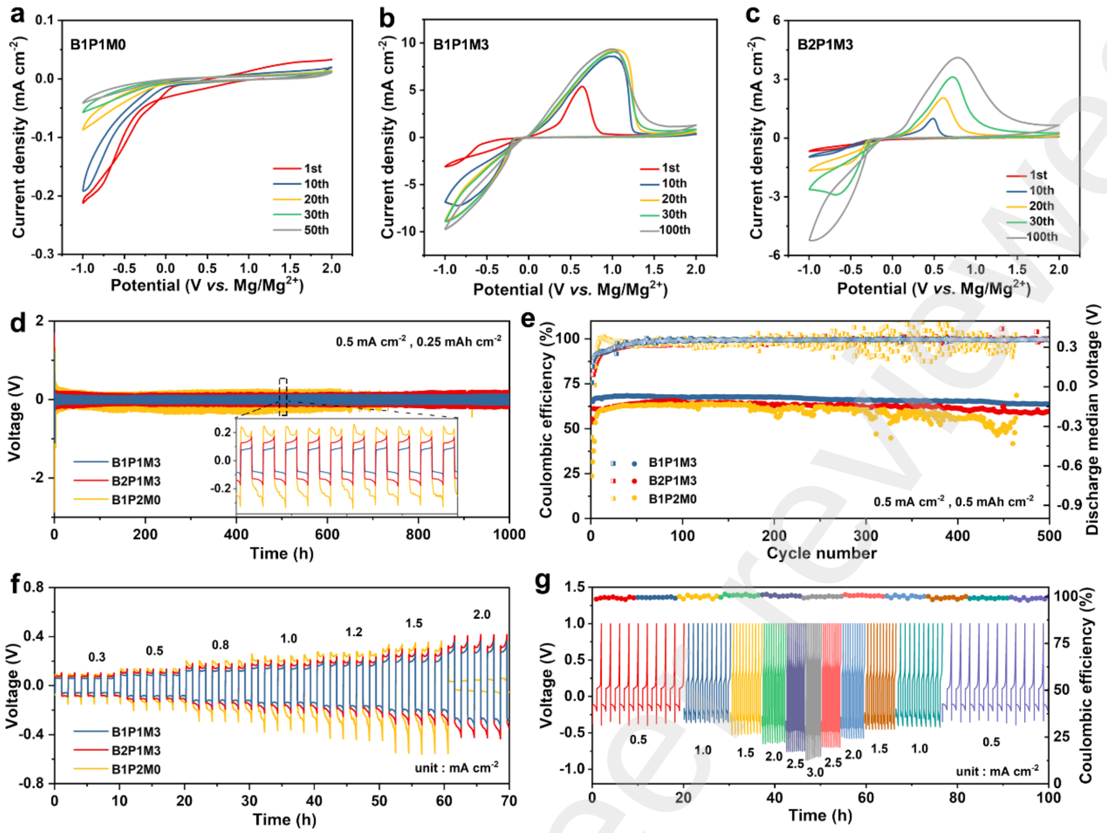

Figure 3. Electrochemical properties of MPFBC electrolytes. CV curves of different electrolytes on SS electrode (a) B1P1M0; (b) B1P1M3; (c) B2P1M3. (d) Cycling performance of Mg||Mg symmetric cells with different electrolytes at a current density of 0.5 mA cm^{-2} with 0.25 mAh cm^{-2} for 1000 h. (e) Coulombic efficiency and discharge median voltage of Mg||SS cells with different electrolytes at current density of 0.5 mA cm^{-2} and area capacity of 0.5 mAh cm^{-2}. (f) Polarization properties of the symmetric Mg||Mg cells with different electrolyte at various current densities. (g) Plating/stripping curves of the Mg|B1P1M3|SS cell at current densities from 0.5 to 3.0 mA cm^{-2} with 0.5 mAh cm^{-2}.

The outstanding electrochemical performance of B1P1M3 is inseparable from the support of favorable interfaces. In order to clarify the mechanism by which B1P1M3 affects the interface, we conduct tests and analyses during both resting process (open circuit) and cycling process (galvanostatic charge discharge). Chemical reactions dominate the formation of initial interface during the resting process, and electrochemical reactions dominate the evolution of interface during the cycling process. Mg|B1P1M3|Mg cells are employed for interfacial studies. One group of cells undergoes the resting process for 48 hours and then executes the cycling process, while the other group skips the resting process and directly performs the cycling process. The cells subjected to different test protocols are systematically labeled for easy identification. A label is marked with the keyword 'RestX+Yth' which means the corresponding cell has been rested for X hours and then run for Y cycles.

Electrochemical impedance spectroscopy (EIS) is used to investigate electrode process and interfacial properties. The Nyquist plots are shown in Figure S24 and Figure S25. We further adapt distribution of relaxation times (DRT) to extract timescale characteristics from Nyquist plots while avoiding the utilization of empirical equivalent circuit in the conventional analysis [39,40]. In Figure 4, DRT plots show four peaks/regions within the time domain, each corresponding to fundamental steps of electrode process. Generally, the order of peak positions (the relaxation time) reflects the sequence of steps, and the peak area represents the resistance of steps. In order of increasing relaxation time, the four peaks/regions are related to bulk resistance (R_b), surface reaction resistance (R_{sr}), mass transport resistance (R_{mt}) and charge transfer resistance (R_{ct}) [41].

During the resting process, R_{ct} experiences the most pronounced increase over time which signifies that the charge transfer on Mg electrode is growing difficult and becomes the rate-determining step (Figure 4a&d and Table S4). Magnesium metal, owning to relatively low electrode potential, has a spontaneous tendency to react with the electrolyte [42]. So continuous chemical reactions between magnesium metal and B1P1M3 lead to the accumulation of interfacial products that are detrimental to charge transfer. This alterations on interfacial environment make the charge transfer step sluggish and cause an increase in the relaxation time.

The chemical reactions also lead to significant changes in electrode morphology as shown in Figure S26. The magnesium surface is not only covered with uneven, loose and insoluble products, but also exhibits typical corrosion features. Similar corrosion morphologies are commonly found in electrolyte containing chloride ions [43]. These numerous tiny corrosive pits are formed with the assistance of chloride ions in B1P1M3. X-ray photoelectron spectroscopy (XPS) (Figure 5a - d and S27) indicates that the main interfacial products are organic species, such as C-O and C-C, which result from solvent molecule decomposition. There are also minor decomposition products of boron-center anion, including B-F (193.5 eV) and B-Ph (191.4 eV) species [44,45].

Prompted by chemical reactions between magnesium metal and electrolyte, mass transport step in the resting process mainly originates from the movement of solvent molecules and boron-center anions through the organic-rich interface layer towards Mg electrode. However, the continuous accumulation of insoluble reaction products cripples their transportation ability to the magnesium metal, leading to an augmentation in R_{mt}. Surface reaction predominantly stems from the adsorption of solvent molecules and chloride ions on the magnesium metal [46,47]. As the resting time extends, the Mg electrode exposes more corrosive pits which provide additional sites for adsorption, ultimately inducing an increase in R_{sr}. To sum up, during the resting process, electrode reaction is irreversible corrosion reaction of B1P1M3 on magnesium metal assisted by chloride ions.

After resting for 48 hours, the cells undergo constant current discharge/charge test. Electrochemical reactions drive changes of the interface, which in turn influences the electrode process. The DRT plots (Figure 4b) describe that the original single peak of charge transfer step splits into two peaks after one cycle, and the associated relaxation time shifts. These findings imply that the charge transfer process has changed. Electrodeposition of Mg involves two-electron transformation which might be experienced in two steps [42,48]. Hence, the double peaks in the charge transfer process are hypothesized as two reversible subprocesses: $Mg^{2+} + e \leftrightarrow Mg^{+}$ (lnτ around 0.5) and $Mg^{+} + e \leftrightarrow Mg$ (lnτ around 3). As shown in Figure 4e and Table S4, the R_{ct} decreases dramatically in the early stage of cycling, indicating that interfacial environment is progressing towards facilitating the reversible Mg^{2+}/Mg conversion. Changes in the charge transfer step must be accompanied by the transitions in mass transport and surface reaction. The mass transport step after cycling may be transferred into the process in which Mg^{2+} ions travel back and forth through the interface layer to cooperate with the reversible Mg^{2+}/Mg conversion. R_{mt} in the current state can be considered as SEI resistance. The reduction in R_{mt} is conducive to the passage of Mg^{2+}

through the SEI. Further, the formation of SEI shields the adsorption sites for solvent molecules and chloride ions, making adsorption no longer the main factor in the surface reaction. Instead, the solvation/desolvation of Mg^{2+} becomes the dominant role to cooperate with the import and outport of Mg^{2+} from SEI. After 30 cycles, the resistance of each part is basically unchanged, and the interface tends to be stable.

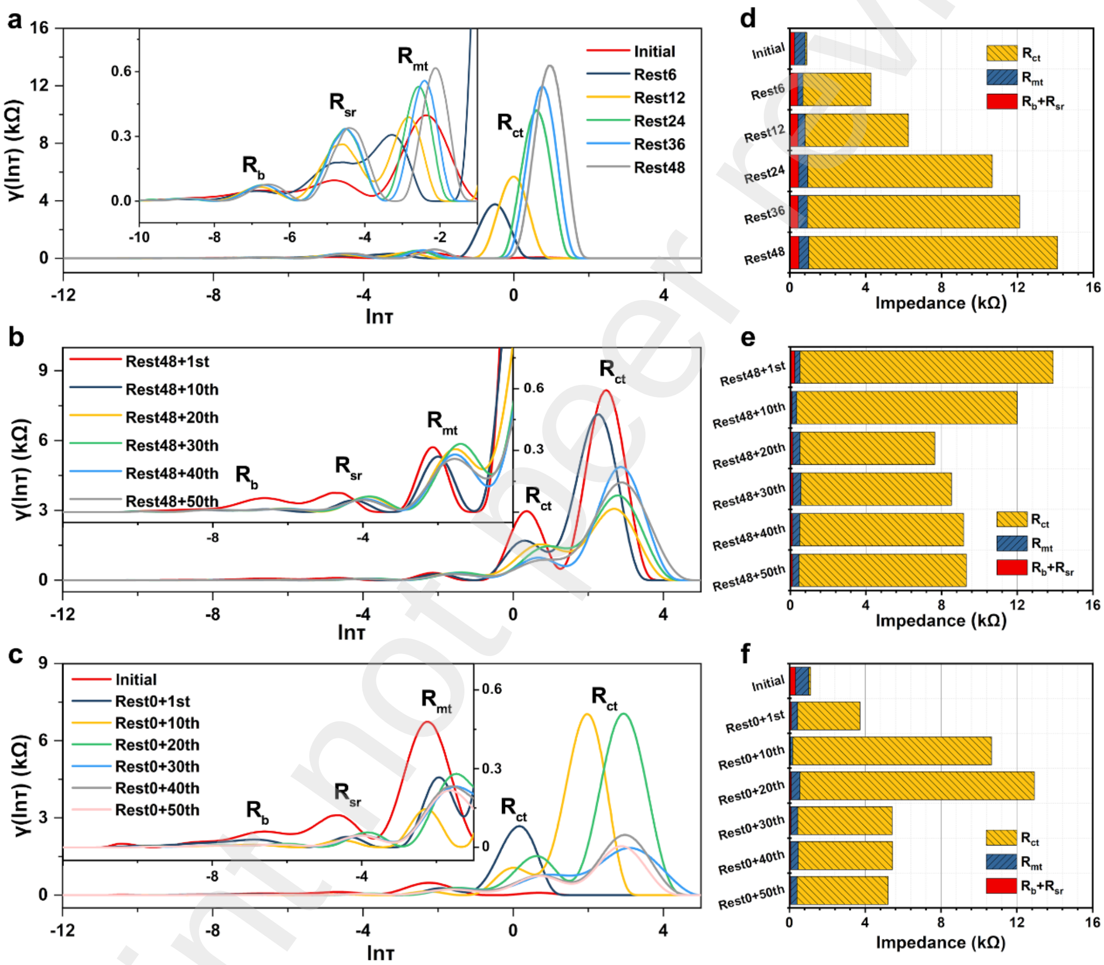

Figure 4. DRT plots of Mg|B1P1M3|Mg cells during (a) the resting process and followed by (b) the cycling process, and (c) the cycling process only. (d-f) Resistance of fundamental steps in the electrode process corresponding to (a-c). The label in figure 'RestX+Yth' indicates the corresponding cell has been rested for X hours and then run for Y cycles.

XPS data of the cycled Mg electrode provides evidence for the tremendous changes mentioned above. The statistical results of atomic proportions show an increase in inorganic constituents within the interface layer after electrochemical process, particularly the concentration of boron-containing species (Figure 5d). The B 1s and F 1s spectra (Figure 5a & 5b and S28 & 29) reveal a simultaneous rise in B-Ph species and MgF_2 (684.7 eV), and this rise may be attributed to the electrochemical decomposition of the phenyl-substituted boron-center anions. For instance, the B-F bond in the Ph_3BF^- anion breaks, resulting in the formation of both B-Ph species and MgF_2. The former is beneficial for the stabilization of interface and the latter is a good Mg^{2+} conductor [7,22]. According to molecular orbital energy levels (Figure 5e), phenyl-substituted boron-center anions have lower LUMO energy levels compared to BF_4^-, which means they are more prone to undergo reduction reactions during Mg plating. The electrochemically reconstructed interface has the characteristics of SEI. It can not only conduct Mg^{2+} ions, but also isolate solvent molecules and anions from contact with magnesium metal, preventing corrosion reactions and the decomposition of the electrolyte. Overall, the electrochemical process could profoundly change the interface chemistry, which shift electrode reaction from corrosion reaction into Mg^{2+}/Mg conversion reaction.

We also monitor Mg||Mg cells that directly perform cycling process without resting process. In freshly assembled cells, R_{ct} is small because of active interface reactions between pristine magnesium and electrolyte. After one cycle, the charge transfer step still maintains a single peak (Figure 4c), which is different from the

Rest48+1st case. XPS data (Figure 5d and S30) reveal that both solvent molecules and boron-center anions decompose, thereby forming the organic and inorganic composite layer. But this immature interface can not prevent the corrosion reaction, which may be attributed to the insufficient inorganic components and lack of robustness. The electrochemically driven interface has not been fully formed, and the corrosion reactions dominate the electrode process in this state. After 10 cycles, double peaks related to reversible Mg^{2+}/Mg conversion appear, meaning that the electrochemically constructed SEI has been established. After the subsequent cycles, the interfacial environment undergoes complex changes until the resistance of each step in the electrode process becomes stable (Figure 4f and Table S5). Compared with the cells that experienced the resting process, the cells without resting have smaller R_{ct} because of fewer corrosion products and better kinetics.

Mg electrodes subjected to different testing protocols exhibit intriguing morphology difference. For the cells that directly undergo the cycling process, Mg^{2+} ions are stripped from the Mg anode during the first cycle, leading to the formation of initial stripping pits (Figure 5h). Throughout repeated cycling, the pits gradually grow larger and form deep, swirling morphologies, while regions outside the pits retain a relatively flat appearance (Figure 5i). This is due to the fact that electrochemical reactions usually tend to occur within pit regions, causing a locally high density of ions and electrons [49,50]. In contrast, for the cells that have experienced the resting process, the surface of Mg anode inherently features numerous pits generated by corrosion, each serving as an active site for Mg^{2+} deposition and dissolution. They can provide a fairly even Mg^{2+} flux and disperse the capacity for Mg stripping/plating, resulting in relatively uniform electrode morphologies with smaller and shallower pits after cycling. As depicted in Figure 5f - i, the pits observed in the rest48+1st and rest48+50th are smaller than those seen in the rest0+1st and rest0+50th.

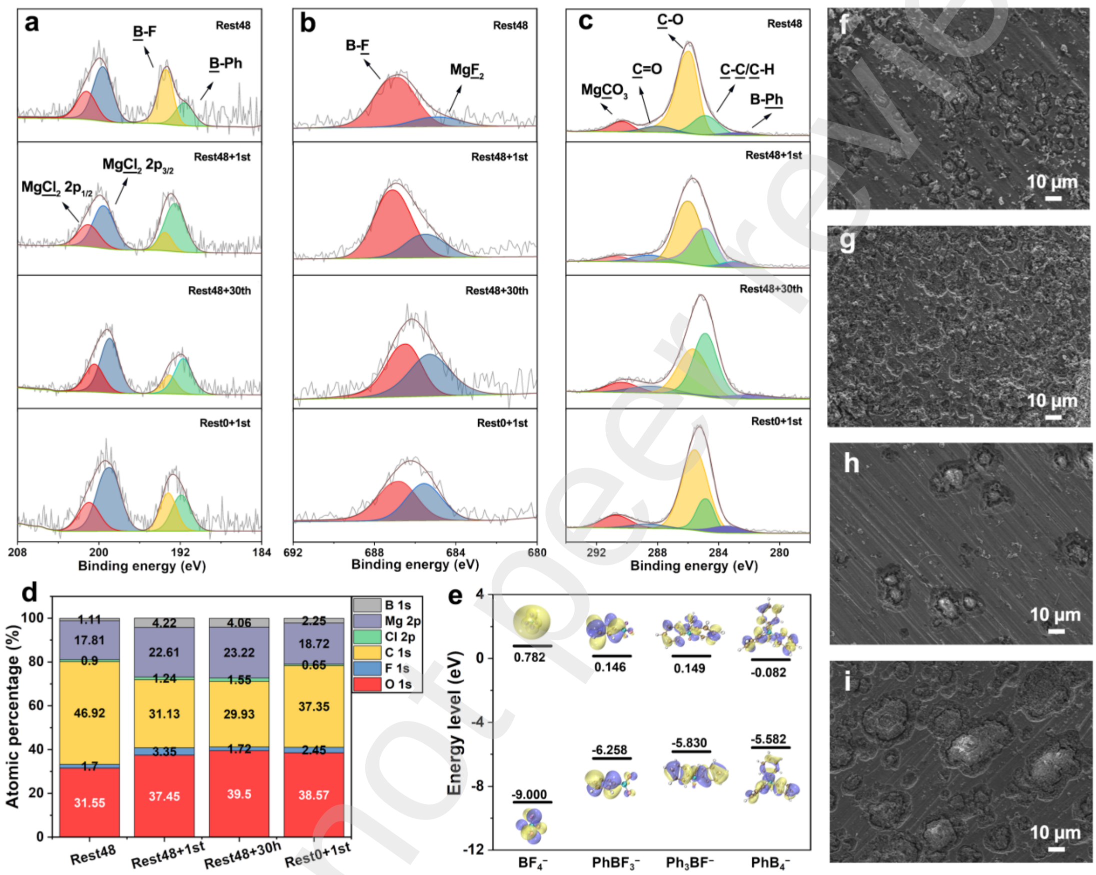

Figure 5. XPS analysis of (a) Cl 2p and B 1s, (b) F 1s, (d) C 1s on Mg electrode under different states. (d) Atomic proportions of interface layer on Mg electrode. (e) LUMO and HOMO energy levels of Ph_4B^-, Ph_3BF^-, $PhBF_3^-$ and BF_4^- anions. SEM images of the cycled Mg electrode (f) Rest48+1st, (g) Rest48+50th, (h) Rest0+1st and (i) Rest0+50th.

Additionally, we follow the same procedure to test and analyze the electrode process of the Mg anode in APC electrolyte by EIS-DRT. The results (Figure S31 - 34 and Table S6 & 7) show that the electrode process exhibits characteristics similar to those described in B1P1M3, indicating that the interface evolution is universal, especially in chlorine-containing electrolytes.

Based on in-depth interface characterization, we summarize a comprehensive illustration of the Mg|MPFBC interface evolution mechanism in Figure 6. Upon contact with fresh magnesium metal, the electrolyte components, especially solvent molecules and chloride ions quickly establish an inner Helmholtz layer on the magnesium surface. On the one hand (Path I), during the chemical process, magnesium reacts with the electrolyte, forming numerous small corrosive pits on its surface with the assistance of chloride ions. The reaction products accumulate at the surface, which is not conducive to the mass transport and charge transfer steps, thereby weakening the corrosion reactions. It can be considered that the corrosion-induced interface acts as a passivation layer. Followed by the cycling process, electrochemical reactions occur between the electrolyte and the Mg electrode, generating a B & F-containing interface which can conduct Mg^{2+} and prevent the corrosion reaction. This electrochemistry-driven interface can be regarded as a SEI layer, ensuring both interface stability and reversible Mg^{2+}/Mg conversion. The extensive corrosive pits can disperse the capacity of Mg stripping and plating, resulting in relatively flat Mg electrode morphologies. On the other hand (Path II), the Mg electrode directly executes electrochemical process, almost avoiding the influence of the corrosion. Stripping pits with a small number and large size are formed after the first cycle. In subsequent cycles, the deposition and dissolution of Mg^{2+} concentrate on these pits, leading to larger and deeper pits on the electrode.

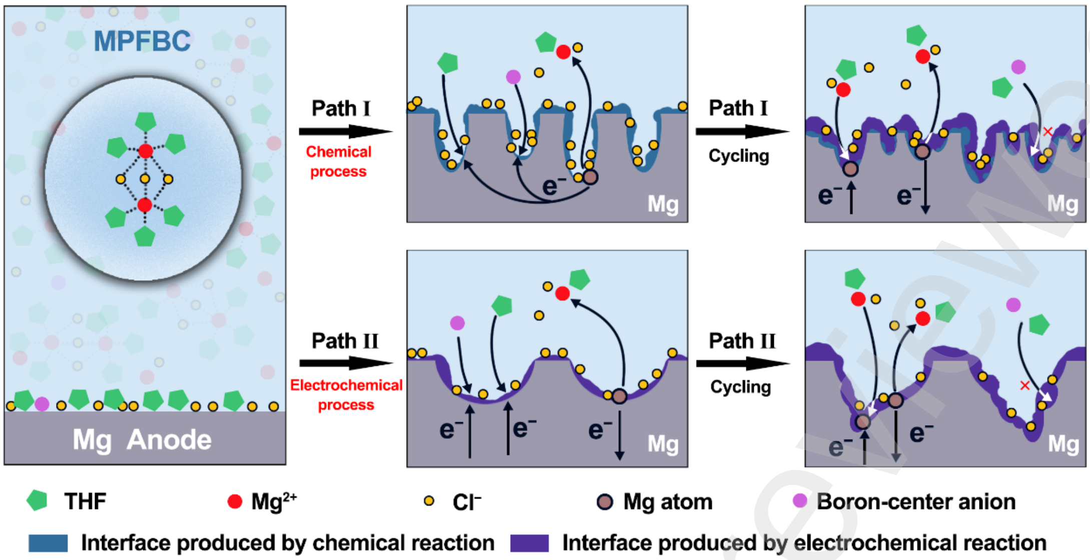

Figure 6. Schematic illustrations for the interface evolution of Mg anode in MPFBC.

In summary, the corrosion pits provided by the chemical process compensate for the deficiencies of stripping pits from the electrochemical process, while the activation effect of the electrochemical process rescues the passive interface formed by the chemical process. The chemistry-electrochemistry complementary advantages can be reasonably leveraged to regulate interface chemistry and interface evolution of Mg anode in chlorine-containing electrolytes as Path I: initially obtaining a large number of tiny corrosion pits through chemical process, and then constructing the SEI via electrochemical process. This methodology not only ensures a stable and Mg^{2+}-conducted interface, but also provides a precondition for developing uniform electrode morphology.

3. Conclusions

In conclusion, an innovative and accessible fluorinated boron-based electrolyte

MPFBC has been designed and synthesized, utilizing cheap raw material BF_3 and a facile Lewis acid-base reaction. The electrolyte comprises phenyl-substituted boron-center anions that construct a B & F-containing interface through chemical and electrochemical reactions, which is crucial for achieving reversible Mg^{2+} deposition/dissolution and stable battery cycling. MPFBC can support stable cycling of Mg||Mg cells at 0.5 mA cm^{-2} with low overpotential (<100 mV) and maintain a decent discharge specific capacity of 62.6 mAh g^{-1} after 1000 cycles of Mg||Mo_6S_8 battery at 0.5 C. In-depth interface research reveals that the chemical process accompanying the corrosion reaction can passivate Mg anode, but the formed corrosion pits are the initial morphology conducive to dispersing the Mg^{2+} flux and promoting uniform Mg stripping/plating during cycling. Conversely, the electrochemical process can activate the Mg anode, but favorable initial morphology cannot be formed. Utilizing the complementary advantages of corrosive morphology induced by chemical processes and SEI constructed through electrochemical processes could regulate interface chemistry and interface evolution. This work provides a boron-based electrolyte which is suitable for large-scale application and takes both price and performance into account, and contributes a practical and universal interface regulation strategy for RMBs.

4. Experimental Section

4.1 Materials and Chemicals

Boron trifluoride tetrahydrofuran complex (the content of BF_3 is approximately 48-50% titrated by NaOH), phenylmagnesium chloride (2.0 M PhMgCl in tetrahydrofuran), magnesium chloride ($MgCl_2$, >98%) and aluminum chloride ($AlCl_3$, 99.5%) were obtained from Aladdin. Anhydrous tetrahydrofuran (THF, 99.5%) was purchased from Macklin. All chemicals and reagents were used directly without purification.

4.2 Preparation of Electrolytes

All operations for the electrolyte were carried out in an Ar-filled glove box with H_2O and O_2 below 1 ppm. The preparation of the MPFBC electrolyte involved the following steps. First, boron trifluoride tetrahydrofuran complex and phenylmagnesium chloride were individually diluted with THF. Then, those two solutions were mixed and stirred for at least 6 h. Subsequently, $MgCl_2$ was selectively added to the above mixture and stirred at 35 ^oC for more than 18 h to obtain the transparent electrolyte.

All-phenyl complex (APC) electrolyte was prepared according to a literature procedure [30]. Briefly, 0.32 g of $AlCl_3$ was slowly added into 3.6 mL of THF with continuous stirring until it was fully dissolved. Then, 2.4 mL of 2.0 M PhMgCl solution was slowly added into the above solution, followed by stirring for 12 h.

4.3 Electrochemical Measurements

All electrochemical measurements were conducted on a CHI 660E electrochemical workstation (Shanghai, China) or a Neware battery test system (Shenzhen, China) at 30 ^oC. Unless otherwise indicated, cells for performance assessment were assembled using a 2032-type coin cell with 120 μL of electrolyte and a piece of glass microfiber separator (Whatman, GF/A).

Cyclic voltammetry (CV) was performed using Mg||SS cells at a scan rate of 25 mV s^{-1}. The oxidation stability of the electrolytes was examined on various working electrodes (Pt, Mo and SS foils) using linear sweep voltammetry (LSV) at a scan rate of 5 mV s^{-1} in a three-electrode system. The Mg foil served as both the reference and counter electrode in this setup. The ionic conductivity of electrolytes was measured by electrochemical impedance spectroscopy (EIS) using two SS blocking electrodes [51]. Symmetric Mg||Mg cells were fabricated to evaluate the polarization properties and long-cycle performance of electrolytes. Asymmetric Mg||SS cells were assembled to verify the coulombic efficiency (CE) of electrolyte.

Mg||Mg cells were assembled for investigating interfacial properties by EIS. It was conducted within the range of 0.1 MHz to 10 mHz with amplitude of 5 mV. For the sake of test consistency, the EIS is carried out two minutes after the completion of battery charging and discharging. Open-source MATLAB script-based software (DRT Tools) was used to calculate the distribution of relaxation times (DRT) from the impedance data [52]. The Tikhonov regularization was used to fit discrete experimental data in a non-linear least-square manner and the Gaussian method was used for data discretization. Second-order regularization derivative fitting parameters were used and the regularization parameter was set at 0.0001, while radial basis function (RBF) with FWHM of 0.5 was used.

4.4 Characterization methods

The active species of the electrolyte were investigated with nuclear magnetic resonance (NMR, Bruker AVANCE600), electrospray ionization-mass spectrometry (ESI-MS, Agilent QTOF6550) and Raman spectrometry (LabRAM HR Evolution). The ^{11}B and ^{19}F NMR spectra were obtained by testing a mixture consisting of 0.5 mL of electrolyte and 50 μL of deuterated chloroform. ESI-MS was recorded by using CH_3OH as the mobile phase. Raman shifts were measured by using a He-Ne laser (532 nm) in sealed quartz cuvettes.

Scanning electron microscopy (SEM) was performed with a field emission scanning electron microscope (FESEM, JEOL JSM-7800F). X-ray photoelectron spectroscopy (XPS, Thermo Scientific K-Alpha+) with Al Kα radiation as the X-ray source was performed to investigate the chemical composition of the electrode surface. Depth profiles from top to bottom were collected after argon sputtering. It was noted that a polypropylene (PP) separator was inserted between the glass fiber separator and the Mg electrode to prevent the glass fiber from adhering to the electrode.

4.5 Fabrication of cells

Mg||Mo_6S_8 and Mg||CuS cells were assembled with 2032 coin-type configuration to evaluate the practical battery performance of electrolytes. Mo_6S_8 and CuS were prepared according to previous reports [53,54]. The cathodes were prepared by mixing 70% Mo_6S_8 or CuS, 20% Super P and 10% poly(vinylidene fluoride) (PVDF) in N-methylpyrrolidinone (NMP) to form a slurry using a mortar and pestle. The resulting slurry was spread on SS foil and dried under vacuum at 60 °C for 24 h. Polished Mg foils (Φ14 mm, 50 μm) were used as anodes, and Mo_6S_8 (Φ10 mm) or CuS (Φ12 mm) were used as cathodes. The areal density of Mo_6S_8 and CuS cathodes was around 1.0-1.2 mg cm^{-2}.

CRediT authorship contribution statement

Xiaochen Liu: Conceptualization, Methodology, Visualization, Data curation, Writing – original draft. **Jiaxin Wen:** Resources, Formal analysis, Supervision. **Xingwang Zhao:** Methodology, Formal analysis. **Bo Shang:** Software, Validation. **Jinglei Lei:** Validation, Writing – review and editing. **Lingjie Li:** Conceptualization, Project administration, Funding acquisition, Writing – review & editing. **Fusheng Pan:** Resources.

Acknowledgements

This work was supported by the National Key Research and Development Program of China (No. 2023YFB3809500), the Graduate Scientific Research and Innovation Foundation of Chongqing (No. CYB23013), the Fundamental Research Funds for the Central Universities (No. 2024CDJXY010), the Doctoral Research Foundation of Chongqing Industry Polytechnic College (No. 2022GZYBSZK2-11).

Appendix A. Supplementary material

Supplementary data associated with this article can be found in the online version.

Data Availability

Data will be made available on request.

References

[1] C. Senthil, C.W. Lee, Biomass-derived biochar materials as sustainable energy sources for electrochemical energy storage devices, Renew. Sust. Energ. Rev. 137 (2021) 110464. https://doi.org/10.1016/j.rser.2020.110464.
[2] P. Liu, J. Long, R. Wang, Y. Zhou, B. Qu, L. Zhang, X. Zhou, Q. An, Fluorine-based localization effect for stabilized high-voltage magnesium phenolic electrolyte, Energy Storage Mater. 71 (2024) 103679. https://doi.org/https://doi.org/10.1016/j.ensm.2024.103679.
[3] D. Aurbach, Z. Lu, A. Schechter, Y. Gofer, H. Gizbar, R. Turgeman, Y. Cohen, M. Moshkovich, E. Levi, Prototype systems for rechargeable magnesium batteries, Nature 407 (2000) 724-727. https://doi.org/10.1038/35037553.
[4] S. Tan, J. Xu, R. Deng, Q. Zhao, C. Xu, G. Huang, J. Wang, F. Pan, A perspective on the key factors of safety for rechargeable magnesium batteries, J. Energy Chem. 94 (2024) 656-676. https://doi.org/https://doi.org/10.1016/j.jechem.2024.03.015.

[5] H. Zhang, L. Qiao, M. Armand, Organic Electrolyte Design for Rechargeable Batteries: From Lithium to Magnesium, Angew. Chem. Int. Ed. 61 (2022) e202214054. https://doi.org/10.1002/anie.202214054.
[6] R. Mohtadi, O. Tutusaus, T.S. Arthur, Z. Zhao-Karger, M. Fichtner, The metamorphosis of rechargeable magnesium batteries, Joule 5 (2021) 581-617. https://doi.org/10.1016/j.joule.2020.12.021.
[7] H. Dou, X. Zhao, Y. Zhang, W. Zhao, Y. Yan, Z.-F. Ma, X. Wang, X. Yang, Revisiting the degradation of solid/electrolyte interfaces of magnesium metal anodes: Decisive role of interfacial composition, Nano Energy 86 (2021) 106087. https://doi.org/10.1016/j.nanoen.2021.106087.
[8] H. Xiao, X. Li, Y. Fu, Advances in Anion Chemistry in the Electrolyte Design for Better Lithium Batteries, Nano-Micro Lett. 17 (2025) 149. https://doi.org/10.1007/s40820-024-01629-5.
[9] M. Cheng, Y. Wang, D. Zhang, S. Zhang, Y. Yang, X. Lv, J. Wang, Y. NuLi, An efficient Hauser-base electrolyte for rechargeable magnesium batteries, J. Energy Chem. 76 (2023) 1-10. https://doi.org/10.1016/j.jechem.2022.08.037.
[10] D. Zhang, S. Duan, X. Liu, Y. Yang, Y. Zhang, W. Ren, S. Zhang, M. Cheng, W. Yang, J. Wang, Y. NuLi, Deeping insight of Mg(CF3SO3)2 and comprehensive modified electrolyte with ionic liquid enabling high-performance magnesium batteries, Nano Energy 109 (2023) 108257. https://doi.org/10.1016/j.nanoen.2023.108257.
[11] A. Du, Z. Zhang, H. Qu, Z. Cui, L. Qiao, L. Wang, J. Chai, T. Lu, S. Dong, T. Dong, H. Xu, X. Zhou, G. Cui, An efficient organic magnesium borate-based electrolyte with non-nucleophilic characteristics for magnesium–sulfur battery, Energy Environ. Sci. 10 (2017) 2616-2625. https://doi.org/10.1039/C7EE02304A.
[12] Z. Zhang, Z. Cui, L. Qiao, J. Guan, H. Xu, X. Wang, P. Hu, H. Du, S. Li, X. Zhou, S. Dong, Z. Liu, G. Cui, L. Chen, Novel Design Concepts of Efficient Mg-Ion Electrolytes toward High-Performance Magnesium-Selenium and Magnesium-Sulfur Batteries, Adv. Energy Mater. 7 (2017) 1602055. https://doi.org/10.1002/aenm.201602055.
[13] Y.-s. Guo, F. Zhang, J. Yang, F.-f. Wang, Y. NuLi, S.-i. Hirano, Boron-based electrolyte solutions with wide electrochemical windows for rechargeable magnesium batteries, Energy Environ. Sci. 5 (2012) 9100-9106. https://doi.org/10.1039/c2ee22509c.
[14] L. Ma, J. Tan, Y. Wang, Z. Liu, Y. Yang, T. Gray, X. Zhang, M. Ye, J. Shen, Boron-Based High-Performance Lithium Batteries: Recent Progress, Challenges, and Perspectives, Adv. Energy Mater. 13 (2023) 2300042. https://doi.org/10.1002/aenm.202300042.
[15] P. Jankowski, Z. Li, Z. Zhao-Karger, T. Diemant, M. Fichtner, T. Vegge, J.M.G. Lastra, Development of Magnesium Borate Electrolytes: Explaining the Success of $Mg[B(hfip)_4]_2$ Salt, Energy Storage Mater. 45 (2022) 1133-1143. https://doi.org/10.1016/j.ensm.2021.11.012.
[16] Z. Zhao-Karger, R. Liu, W. Dai, Z. Li, T. Diemant, B.P. Vinayan, C.B. Minella, X. Yu, A. Manthiram, R.J. Behm, M. Ruben, M. Fichtner, Toward Highly Reversible Magnesium-Sulfur Batteries with Efficient and Practical $Mg[B(hfip)_4]_2$ Electrolyte, ACS Energy Lett. 3 (2018) 2005-2013. https://doi.org/10.1021/acsenergylett.8b01061.
[17] M. Cheng, W. Ren, D. Zhang, S. Zhang, Y. Yang, X. Lv, J. Yang, J. Wang, Y. NuLi, Efficient single-perfluorinated borate-based electrolytes for rechargeable magnesium batteries, Energy Storage Mater. 51 (2022) 764-776. https://doi.org/10.1016/j.ensm.2022.07.021.
[18] W. Ren, D. Wu, Y. NuLi, D. Zhang, Y. Yang, Y. Wang, J. Yang, J. Wang, An Efficient Bulky $Mg[B(Otfe)_4]_2$ Electrolyte and Its Derivatively General Design Strategy for Rechargeable Magnesium Batteries, ACS Energy Lett. 6 (2021) 3212-3220. https://doi.org/10.1021/acsenergylett.1c01411.

[19] J. Luo, Y. Bi, L. Zhang, X. Zhang, T.L. Liu, A Stable, Non-Corrosive Perfluorinated Pinacolatoborate Mg Electrolyte for Rechargeable Mg Batteries, Angew. Chem. Int. Ed. 58 (2019) 6967-6971. https://doi.org/10.1002/anie.201902009.

[20] K. Tang, A. Du, S. Dong, Z. Cui, X. Liu, C. Lu, J. Zhao, X. Zhou, G. Cui, A Stable Solid Electrolyte Interphase for Magnesium Metal Anode Evolved from a Bulky Anion Lithium Salt, Adv. Mater. 32 (2020) 1904987. https://doi.org/10.1002/adma.201904987.

[21] X. Huang, S. Tan, J. Chen, Z. Que, R. Deng, J. Long, F. Xiong, G. Huang, X. Zhou, L. Li, J. Wang, L. Mai, F. Pan, Asymmetric $SO_3CF_3^-$-Grafted Boron-Center Anion Enables Boron-Containing Interphase for High-Performance Rechargeable Mg Batteries, Adv. Funct. Mater. 34 (2024) 2314146. https://doi.org/10.1002/adfm.202314146.

[22] Y. Wang, Y. Sun, D. Zhang, M. Pan, Y. Chen, S. Chen, S. Zhang, Y. Zhao, J. Wang, Y. NuLi, A Hauser-base modulated boron-based electrolyte empowering superior interfacial chemistry in rechargeable magnesium batteries, Energy Storage Mater. 65 (2024) 103152. https://doi.org/10.1016/j.ensm.2023.103152.

[23] J. Long, Y. An, Z. Yang, G. Zhang, J. Zhang, S. Tan, Q. An, Efficient boron-based electrolytes constructed by anionic and interfacial co-regulation for rechargeable magnesium batteries, Chem. Eng. J. 461 (2023) 141901. https://doi.org/10.1016/j.cej.2023.141901.

[24] C. Yan, R. Xu, Y. Xiao, J.-F. Ding, L. Xu, B.-Q. Li, J.-Q. Huang, Toward Critical Electrode/Electrolyte Interfaces in Rechargeable Batteries, Adv. Funct. Mater. 30 (2020) 1909887. https://doi.org/10.1002/adfm.201909887.

[25] Y.-X. Yao, X. Chen, C. Yan, X.-Q. Zhang, W.-L. Cai, J.-Q. Huang, Q. Zhang, Regulating Interfacial Chemistry in Lithium-Ion Batteries by a Weakly Solvating Electrolyte, Angew. Chem. Int. Ed. 60 (2021) 4090-4097. https://doi.org/10.1002/anie.202011482.

[26] X.-C. Hu, Y. Shi, S.-Y. Lang, X. Zhang, L. Gu, Y.-G. Guo, R. Wen, L.-J. Wan, Direct insights into the electrochemical processes at anode/electrolyte interfaces in magnesium-sulfur batteries, Nano Energy 49 (2018) 453-459. https://doi.org/10.1016/j.nanoen.2018.04.066.

[27] T. Wen, H. Xiao, S. Tan, X. Huang, B. Qu, L. Cao, G. Huang, J. Song, J. Wang, A. Tang, J. Yue, F. Pan, Interfacial chemistry of anode/electrolyte interface for rechargeable magnesium batteries, J. Magnes. Alloys 12 (2024) 2647-2673. https://doi.org/https://doi.org/10.1016/j.jma.2024.04.010.

[28] M. Liu, L.-J. Hu, Z.-K. Guan, T.-L. Chen, X.-Y. Zhang, S. Sun, R. Shi, P. Jing, P.-F. Wang, Tailoring Cathode–Electrolyte Interface for High-Power and Stable Lithium–Sulfur Batteries, Nano-Micro Lett. 17 (2024) 85. https://doi.org/10.1007/s40820-024-01573-4.

[29] D. Aurbach, H. Gizbar, A. Schechter, O. Chusid, H.E. Gottlieb, Y. Gofer, I. Goldberg, Electrolyte solutions for rechargeable magnesium batteries based on organomagnesium chloroaluminate complexes, J. Electrochem. Soc. 149 (2002) A115-A121. https://doi.org/10.1149/1.1429925.

[30] O. Mizrahi, N. Amir, E. Pollak, O. Chusid, V. Marks, H. Gottlieb, L. Larush, E. Zinigrad, D. Aurbach, Electrolyte Solutions with a Wide Electrochemical Window for Rechargeable Magnesium Batteries, J. Electrochem. Soc. 155 (2008) A103. https://doi.org/10.1149/1.2806175.

[31] V. Jonas, G. Frenking, M.T. Reetz, Comparative Theoretical Study of Lewis Acid-Base Complexes of BH_3, BF_3, BCl_3, $AlCl_3$, and SO_2, J. Am. Chem. Soc. 116 (1994) 8741-8753. https://doi.org/10.1021/ja00098a037.

[32] J.I. Martínez-Araya, The dual descriptor potential, J. Math. Chem. 62 (2024) 1094-1112. https://doi.org/10.1007/s10910-024-01583-y.

[33] M. Gacitúa, A. Carreño, R. Morales-Guevara, D. Páez-Hernández, J.I. Martínez-Araya, E. Araya,

M. Preite, C. Otero, M.M. Rivera-Zaldívar, A. Silva, J.A. Fuentes, Physicochemical and Theoretical Characterization of a New Small Non-Metal Schiff Base with a Differential Antimicrobial Effect against Gram-Positive Bacteria, Int. J. Mol. Sci. 23 (2022) 2553. https://doi.org/10.3390/ijms23052553.

[34] N. Pour, Y. Gofer, D.T. Major, D. Aurbach, Structural Analysis of Electrolyte Solutions for Rechargeable Mg Batteries by Stereoscopic Means and DFT Calculations, J. Am. Chem. Soc. 133 (2011) 6270-6278. https://doi.org/10.1021/ja1098512.

[35] Y. Sun, F. Ai, Y.-C. Lu, Electrolyte and Interphase Design for Magnesium Anode: Major Challenges and Perspectives, Small 18 (2022) 2200009. https://doi.org/10.1002/smll.202200009.

[36] B. Pan, J. Huang, N. Sa, S.M. Brombosz, J.T. Vaughey, L. Zhang, A.K. Burrell, Z. Zhang, C. Liao, $MgCl_2$: The Key Ingredient to Improve Chloride Containing Electrolytes for Rechargeable Magnesium-Ion Batteries, J. Electrochem. Soc. 163 (2016) A1672-A1677. https://doi.org/10.1149/2.0821608jes.

[37] K.A. See, K.W. Chapman, L. Zhu, K.M. Wiaderek, O.J. Borkiewicz, C.J. Barile, P.J. Chupas, A.A. Gewirth, The Interplay of Al and Mg Speciation in Advanced Mg Battery Electrolyte Solutions, J. Am. Chem. Soc. 138 (2016) 328-337. https://doi.org/10.1021/jacs.5b10987.

[38] T. Liu, J.T. Cox, D. Hu, X. Deng, J. Hu, M.Y. Hu, J. Xiao, Y. Shao, K. Tang, J. Liu, A fundamental study on the $(\mu\text{-}Cl)_3Mg_2(THF)_6^+$ dimer electrolytes for rechargeable Mg batteries, Chem. Commun. 51 (2015) 2312-2315. https://doi.org/10.1039/c4cc07621d.

[39] X. Chen, L. Li, M. Liu, T. Huang, A. Yu, Detection of lithium plating in lithium-ion batteries by distribution of relaxation times, J. Power Sources 496 (2021) 229867. https://doi.org/10.1016/j.jpowsour.2021.229867.

[40] J. Chen, E. Quattrocchi, F. Ciucci, Y. Chen, Charging processes in lithium-oxygen batteries unraveled through the lens of the distribution of relaxation times, Chem 9 (2023) 2267-2281. https://doi.org/10.1016/j.chempr.2023.04.022.

[41] V. Prabhakaran, R. Kalsar, L. Strange, O.A. Marina, R. Prabhakaran, V.V. Joshi, Understanding Localized Corrosion on Metal Surfaces Using Scanning Electrochemical Cell Impedance Microscopy (SECCIM), J. Phys. Chem. C 126 (2022) 12519-12526. https://doi.org/10.1021/acs.jpcc.2c03807.

[42] H.D. Yoo, I. Shterenberg, Y. Gofer, G. Gershinsky, N. Pour, D. Aurbach, Mg rechargeable batteries: an on-going challenge, Energy Environ. Sci. 6 (2013) 2265-2279. https://doi.org/10.1039/c3ee40871j.

[43] Z. Cui, F. Ge, Y. Lin, L. Wang, L. Lei, H. Tian, M. Yu, X. Wang, Corrosion behavior of AZ31 magnesium alloy in the chloride solution containing ammonium nitrate, Electrochim. Acta 278 (2018) 421-437. https://doi.org/10.1016/j.electacta.2018.05.059.

[44] W. Xue, Y. Qin, F. Li, Y. Wang, Z. Wang, Preparation of Ru- bmim BF_4 Catalyst Using $NaBH_4$ as Reducing Agent and Its Performance in Selective Hydrogenation of Benzene, Chin. J. Catal. 33 (2012) 1913-1918. https://doi.org/10.1016/s1872-2067(11)60469-5.

[45] R.L. Tapping, Boron 1s binding energies in MBPh4, where M is sodium, potassium, rubidium, cesium and ammonium, Inorg. Chem. 21 (1982) 1691-1692. https://doi.org/10.1021/ic00134a089.

[46] W. Liu, F. Cao, A. Chen, L. Chang, J. Zhang, C. Cao, Effect of Chloride Ion Concentration on Electrochemical Behavior and Corrosion Product of AM60 Magnesium Alloy in Aqueous Solutions, Corrosion 68 (2012) 045001. https://doi.org/10.5006/0010-9312-68-4-2.

[47] M.A. Gebbie, B. Liu, W. Guo, S.R. Anderson, S.G. Johnstone, Linking Electric Double Layer Formation to Electrocatalytic Activity, ACS catal. 13 (2023) 16222-16239. https://doi.org/10.1021/acscatal.3c04255.

[48] Y. Viestfrid, M.D. Levi, Y. Gofer, D. Aurbach, Microelectrode studies of reversible Mg deposition in THF solutions containing complexes of alkylaluminum chlorides and dialkylmagnesium, J.

Electroanal. Chem. 576 (2005) 183-195. https://doi.org/10.1016/j.jelechem.2004.09.034.

[49] X. Liu, A. Du, Z. Guo, C. Wang, X. Zhou, J. Zhao, F. Sun, S. Dong, G. Cui, Uneven Stripping Behavior, an Unheeded Killer of Mg Anodes, Adv. Mater. 34 (2022) 2201886. https://doi.org/10.1002/adma.202201886.

[50] G. Yang, Y. Li, C. Zhang, J. Wang, Y. Bai, C.Y.J. Lim, M.-F. Ng, Z. Chang, S. Kumar, Z. Sofer, W. Liu, Z.W. Seh, In Situ Formed Magnesiophilic Sites Guiding Uniform Deposition for Stable Magnesium Metal Anodes, Nano Lett. 22 (2022) 9138-9146. https://doi.org/10.1021/acs.nanolett.2c03710.

[51] W. He, Z. Cui, X. Liu, Y. Cui, J. Chai, X. Zhou, Z. Liu, G. Cui, Carbonate-linked poly(ethylene oxide) polymer electrolytes towards high performance solid state lithium batteries, Electrochim. Acta 225 (2017) 151-159. https://doi.org/10.1016/j.electacta.2016.12.113.

[52] T.H. Wan, M. Saccoccio, C. Chen, F. Ciucci, Influence of the Discretization Methods on the Distribution of Relaxation Times Deconvolution: Implementing Radial Basis Functions with DRTtools, Electrochim. Acta 184 (2015) 483-499. https://doi.org/10.1016/j.electacta.2015.09.097.

[53] E. Lancry, E. Levi, A. Mitelman, S. Malovany, D. Aurbach, Molten salt synthesis (MSS) of Cu2Mo6S8—New way for large-scale production of Chevrel phases, J. Solid State Chem. 179 (2006) 1879-1882. https://doi.org/10.1016/j.jssc.2006.02.032.

[54] W. Ren, F. Xiong, Y. Fan, Y. Xiong, Z. Jian, Hierarchical Copper Sulfide Porous Nanocages for Rechargeable Multivalent-Ion Batteries, ACS Appl. Mater. Inter. 12 (2020) 10471-10478. https://doi.org/10.1021/acsami.9b21999.